

Sandbox moderate (interpose):

```
if xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ sandbox $((1<<SYS_open)) - cat README
syntax
$ sandbox $((1<<SYS_open)) - cat README
syntax
$ sanbox 32768 - cat README
exec sanbox failed
$ sandbox 32768 - cat README
cat: cannot open README
$ |
```

Sandbox easy:

```
mariu@DESKTOP-0K07MJA: X + v
riscv64-unknown-elf-objdump -t user/_attack | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/attack.sym
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printn-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/secret.o user/secret.printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_secret > user/secret.asm
riscv64-unknown-elf-objdump -t user/_secret | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_lo_forphan user/_dorphan user/_sandbox user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1027 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sandbox 32768 README grep xv6 README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix Version 6 (v6). xv6 loosely follows the structure and style of v6, xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer (kaashoek,rtm@mit.edu). The main purpose of xv6 is as a teaching
$ |
```

Attack:

```
$ QEMU: Terminated
mari@DESKTOP-0K07MJA:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_ll user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_str c user/_zombie user/_logstress user/_forphan user/_dorphan us nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap balloc: first 1033 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/k irtio-mmio.force-legacy=false -drive file=fs.img,if=none,for iver=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ secret MYSecret123
$ attack
MYSecret123
$ |
```

Make grade:

```
mari@DESKTOP-0K07MJA: X + v
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-built ee -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-built ee -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-built ee -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack riscv64-unknown-elf-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kall el/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/ isk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-unknown-elf-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/mari/xv6-labs-2025'
== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (3.2s)
(Old xv6.out.sandbox_mask failure log removed)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (0.6s)
(Old xv6.out.sandbox_fork failure log removed)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (1.2s)
(Old xv6.out.sandbox_path failure log removed)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.7s)
(Old xv6.out.sandbox_most failure log removed)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.1s)
== Test attack ==
$ make qemu-gdb
attack: OK (1.3s)
== Test time ==
time: OK
Score: 45/45
mari@DESKTOP-0K07MJA:~/xv6-labs-2025$ |
```